

Ideation Phase

Customer Problem Statements

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the most suitable crop for my farmland based on the current soil nutrients and environmental conditions.	I am uncertain which crop will produce the highest yield under the existing conditions.	I lack access to expert agricultural advice and data-driven decision support.	Confused and concerned about making an incorrect crop selection that could reduce productivity and income.

PS-2	A farmer	Determine whether a specific crop (e.g., Cotton or Rice) is suitable for cultivation on my land.	I cannot confidently evaluate if my soil and climate conditions meet the crop's requirements.	Assessing crop suitability requires agricultural knowledge and analysis of multiple environmental factors.	Uncertain about investing time and resources in cultivating that crop.
PS-3	An agricultural researcher / agriculture officer	Analyze agricultural datasets to understand crop suitability patterns and provide scientifically supported recommendations to farmers.	It is difficult to manually analyze large datasets containing multiple soil and environmental parameters.	The relationships between nutrients, climate, and crop performance are complex and require machine learning techniques for accurate analysis.	Challenged to provide timely, accurate, and data-driven recommendations for improving agricultural productivity.